

Retail Inventory Analytics Using Python and Google Colab

Abstract

Effective inventory management is critical for retail businesses because it directly impacts product availability, sales performance, customer satisfaction, and profitability. This project focuses on analyzing retail inventory data using Python in Google Colab. The dataset contains information about products, categories, suppliers, stock levels, sales quantities, and pricing details. The objective of the analysis is to identify high-performing product categories, evaluate inventory status, detect products approaching reorder levels, analyze supplier contributions, and generate business recommendations for improving inventory management.

The project utilizes data cleaning, exploratory data analysis (EDA), and data visualization techniques to transform raw inventory data into meaningful business insights. Various charts and statistical analyses were used to support decision-making and identify opportunities for operational improvement.

1. Introduction

Inventory management plays a vital role in retail operations by ensuring that products are available when customers need them while minimizing excess stock and storage costs. Poor inventory management can lead to stockouts, lost sales, overstock situations, and reduced profitability.

The purpose of this project is to perform an end-to-end inventory analysis using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn within the Google Colab environment. The analysis aims to answer key business questions regarding sales performance, stock levels, supplier contributions, and inventory optimization.

Objectives

The primary objectives of the project are:

- Identify product categories generating the highest sales.
- Determine products approaching reorder levels.
- Analyze the relationship between stock availability and sales.
- Evaluate supplier contributions to inventory.
- Generate actionable business recommendations based on inventory performance.

2. Data Understanding and Cleaning

Dataset Overview

The retail inventory dataset contains product-level information including:

i. Product Name	iv. Opening Stock	vii. Units Sold
ii. Product Category	v. Closing Stock	viii. Unit Price
iii. Supplier Name	vi. Reorder Level	ix. Cost Price

The dataset was imported into Google Colab using the Pandas library.

Data Inspection

The following operations were performed:

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|---|--|
| i. Displayed dataset shape and structure. | iii. Generated summary statistics. |
| ii. Examined column names and data types. | iv. Identified missing values and duplicate records. |

Data Cleaning

Several preprocessing steps were performed to improve data quality:

Missing Value Handling

Missing numerical values were identified and replaced using median values to maintain data consistency.

Duplicate Record Check

Duplicate records were checked and removed where necessary to prevent biased analysis.

Stock Validation

Inventory-related columns were validated to ensure no negative stock values existed. This helped maintain data integrity and improve analytical accuracy.

After cleaning, the dataset was considered ready for exploratory analysis.

3. Exploratory Data Analysis

Category-wise Product Distribution

A category-level analysis was performed to determine the number of products available in each category. This helped identify categories with the largest inventory presence.

Category-wise Sales Analysis

Total units sold were aggregated by category to identify the highest-performing product categories. Categories with strong sales performance indicate higher customer demand and revenue potential.

Top-Selling Products

The top ten products based on units sold were identified. These products represent the fastest-moving inventory and contribute significantly to overall sales performance.

Low-Stock Product Analysis

Products with stock levels approaching or below their reorder thresholds were identified. These items require immediate attention to avoid stockouts and potential revenue loss.

Supplier-wise Inventory Distribution

Supplier contributions were analyzed by counting the number of products supplied. This analysis helped identify major suppliers and potential supplier dependency risks.

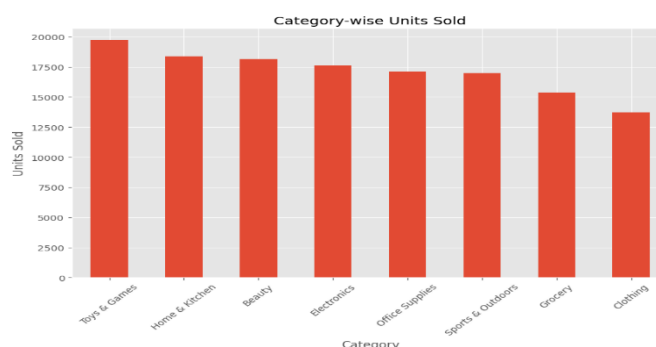
Revenue and Profit Analysis

Additional metrics such as revenue and profit were calculated using pricing and sales information. This enabled identification of products and categories generating the highest profitability.

4. Data Visualization

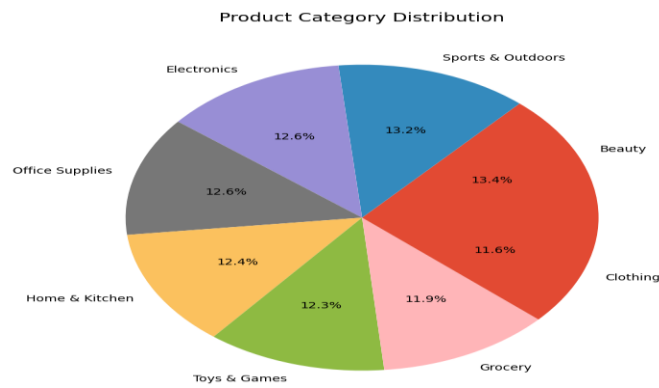
To improve understanding and support decision-making, several visualizations were created.

Bar Chart



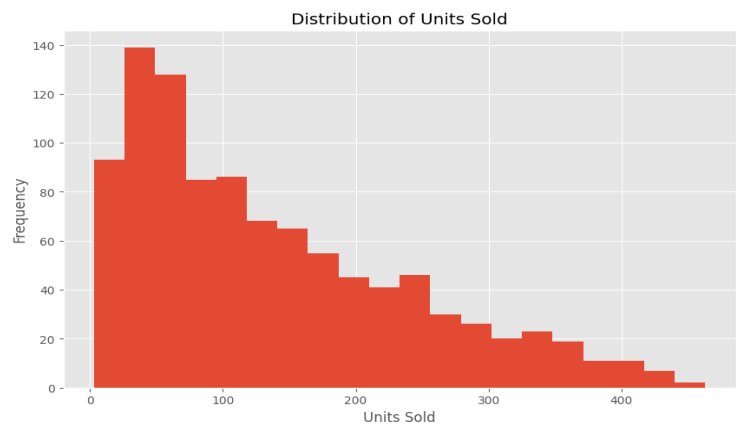
A bar chart was used to display category-wise sales performance. This visualization clearly highlighted the categories generating the highest sales volumes.

Pie Chart



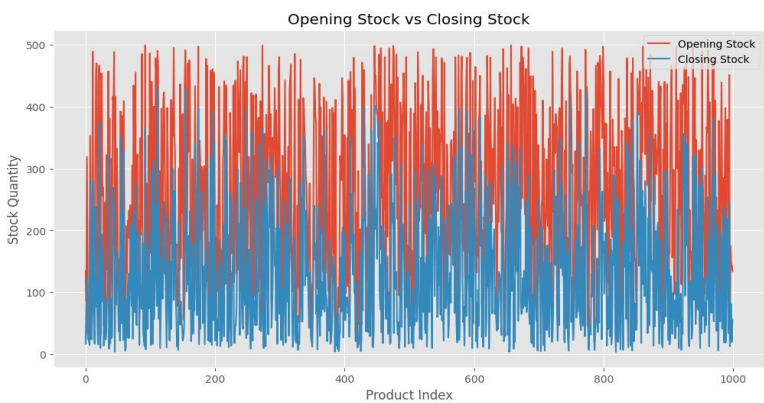
A pie chart illustrated the distribution of products across categories, providing a quick overview of inventory allocation.

Histogram



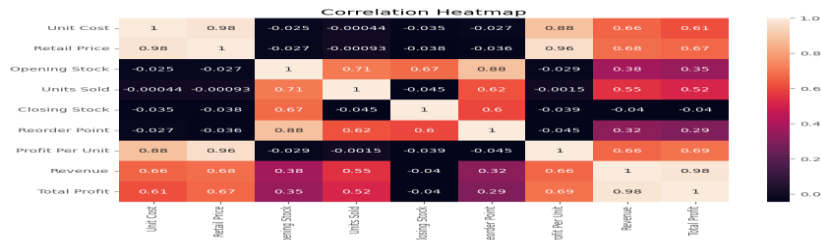
A histogram was used to analyze the distribution of units sold and understand sales frequency patterns across products.

Line Graph



A line graph compared opening stock and closing stock levels, helping visualize inventory movement over time.

Correlation Heatmap



A heatmap was created to identify relationships among numerical variables such as stock levels, sales quantities, prices, revenue, and profit. The correlation analysis provided insights into how inventory variables influence business performance.

These visualizations transformed raw numerical data into meaningful insights that support inventory planning and management decisions.

5. Business Insights and Recommendations

Based on the analysis, several important business insights were identified.

Key Insights

1. Products Requiring Immediate Replenishment

Multiple products were found to have closing stock levels near or below their reorder points. These products should be prioritized for restocking to avoid stock shortages.

2. Fast-Moving Product Categories

Certain product categories demonstrated significantly higher sales volumes than others, indicating strong market demand and revenue-generating potential.

3. Slow-Moving Inventory

Several products exhibited low sales despite maintaining high stock levels. These products may contribute to excess inventory carrying costs and require promotional strategies or inventory reduction plans.

4. High-Profit Opportunities

Products with strong sales performance and high profit margins present opportunities for increased investment and marketing focus.

5. Supplier Dependency Risks

A significant proportion of inventory is supplied by a limited number of suppliers. Heavy dependence on a small supplier base may create operational risks if supply disruptions occur.

Recommendations

- Implement automated reorder alerts for low-stock products.
 - Increase procurement for fast-selling categories.
 - Reduce inventory levels for slow-moving products.
 - Prioritize high-profit products in marketing campaigns.
 - Diversify supplier relationships to reduce dependency risks.
 - Continuously monitor inventory turnover using dashboards and analytics.
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Conclusion

This project successfully demonstrated how data analytics can support inventory management decisions in a retail environment. Through data cleaning, exploratory analysis, visualization, and business intelligence techniques, valuable insights were generated regarding sales performance, stock management, supplier contributions, and profitability.

The analysis provides management with actionable recommendations to improve inventory control, reduce stock-related risks, increase profitability, and enhance operational efficiency. The use of Google Colab and Python libraries proved to be an effective solution for performing retail inventory analytics and generating data-driven business insights.